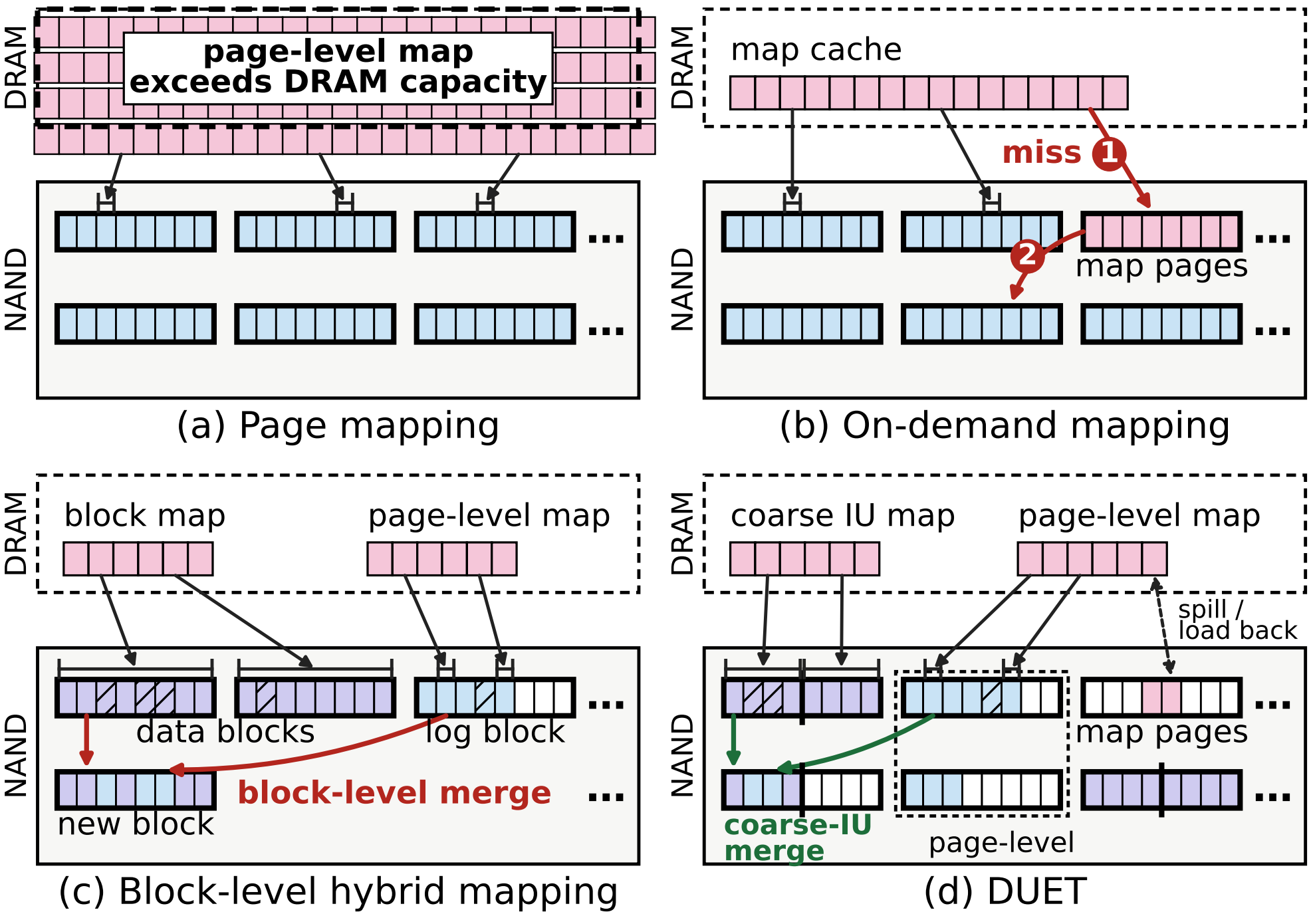
DRAM
page-level map
exceeds DRAM capacity
NAND
(a) Page mapping
DRAM
map cache
miss
1
2
map pages
NAND
(b) On-demand mapping
DRAM
block map
page-level map
data blocks
log block
block-level merge
new block
NAND
(c) Block-level hybrid mapping
DRAM
coarse IU map
page-level map
spill /
load back
map pages
coarse-IU
merge
page-level
NAND
(d) DUET